

Leave Management System

(Product Owner Draft – Scrum Based)

1. Product overview

Leave management system is a web based application designed to optimize the process of leave application, approval, and effective tracking within an organization. It eliminates manual processes and ensures transparency and efficiency.

2. Objective

- To simplify leave application and approval process
- To provide real time leave balance tracking
- To enable efficient communication between employees and managers
- To automate notifications and reporting

3. Stakeholders

- Product Owner
- Development Team
- QA Team
- Employees
- Managers
- Admin

4. Users Roles and Responsibilities

1) Product Owner:

- a) Managing the product backlog
- b) Defining Requirement for Leave Management System
- c) Stakeholder Collaboration

2) Development Team:

- a) Implementing features through code
- b) Delivering software products
- c) Problem solving and optimizing performance
- d)

3) QA Team:

- a) Performing regular automated tests (regression, smoke testing)
- b) Defect tracking and reporting
- c) Perform non-functional testing (performance and security)

4) Employee:

- a) Applying for leave
- b) Viewing leave balance
- c) Tracking leave status

5) Manager:

- a) Approving or rejecting leave
- b) Viewing team leave schedule
- c) Storing reason/comments

6) Admin:

- a) Managing leave policies and ACLs(Access Control List)
- b) Configuring leave types(sick leave, vacation leave)
- c) Generating reports

5. Key Features

- Leave application (full/half day)
- Leave approval workflow
- Leave balance tracking
- Notification system (email/Slack)
- Reporting dashboard

6. Use Cases

TC ID	Scenario	Precondition	Steps	Expected Result
TC-01	Apply for a full day leave with a valid reason.	Employee is logged in and has sufficient leave balance.	1. Navigate to "Apply Leave". 2. Select Leave Type (e.g., Sick, Holiday). 3. Select Dates. 4. Enter Reason. 5. Click "Submit".	Request is submitted successfully; status changes to "Pending Manager Approval".

TC-02	Apply for a half day leave (1st or 2nd half).	Employee is logged in and has sufficient leave balance.	1. Navigate to "Apply Leave". 2. Select date. 3. Check the "Half-Day" option. 4. Select "1st Half" or "2nd Half". 5. Enter Reason and Submit.	System deducts only 0.5 days from the projected balance; request shows as "Pending".
TC-03	Attempt to apply for leave without providing a reason.	Employee is on the "Apply Leave" screen.	1. Fill out dates and leave type. 2. Leave the "Reason" text box empty. 3. Click "Submit".	System blocks submission and displays an error: "Reason is mandatory".
TC-04	Manager reviews and approves a leave request.	Manager is logged in; employee has a pending request.	1. Go to "Pending Approvals". 2. Click on the Employee's request. 3. Review the provided reason. 4. Click "Approve".	Status updates to "Approved"; leave balance is officially deducted.
TC-05	Manager rejects a leave request due to insufficient reason.	Manager is logged in; employee has a pending request.	1. Go to "Pending Approvals". 2. Review the reason. 3. Click "Reject". 4. Enter rejection comments. 5. Click "Confirm".	Status updates to "Rejected"; leave balance is refunded/restored to the employee.
TC-06	Employees track their leave balance and history.	Employee is logged in.	1. Navigate to "My Leave Dashboard". 2. View the summary widget.	Dashboard accurately displays Total Allowed, Used, and Remaining balance for each leave type.

TC-07	Team notification triggers upon manager approval.	Manager has just clicked "Approve" on a leave request.	1. Wait for system processing. 2. Check the team's communication channel (e.g., Email, Slack, Teams).	An automated message is sent to the team: "[Employee Name] will be on leave on [Dates]".
TC-08	Apply for Maternity Leave.	Female employee is logged in and meets tenure criteria.	1. Select "Maternity Leave". 2. Select date range. 3. Upload required medical documents. 4. Click Submit.	System accepts the long-term date range specific to maternity policy and sets status to Pending.
TC-09	Apply for leave on an already booked date.	Employee has an approved leave for Dec 1st.	1. Navigate to "Apply Leave". 2. Select Dec 1st. 3. Submit.	System displays an error: "You already have a leave scheduled for this date."
TC-10	Apply for leave spanning a Public Holiday.	Dec 25th is marked as a Public Holiday in the system.	1. Apply for leave from Dec 24th to Dec 26th.	System detects the holiday and deducts only 2 days (not 3) from balance.
TC-11	Manager provides a rejection reason.	Manager is rejecting a request.	1. Click "Reject". 2. Leave "Comments" empty. 3. Click "Submit".	System requires a rejection comment so the employee knows why it was denied.
TC-12	Verify Team Notification Content.	A leave has been approved.	1. Open the team notification email/alert.	Notification includes: Employee Name, Date Range, and "Half-Day" status if applicable.

7. Scrum Life Cycle

7.1 Product Backlog

Story ID	Title	User Story (Requirement)	Priority
US-01	Leave Type Selection	As an employee, I want to select a leave type (Sick, Holiday, Maternity) so that my leave is categorized correctly.	High
US-02	Compulsory Reason	As a manager, I want to ensure employees provide a reason for their leave so that I can judge the validity of the request.	High
US-03	Manager Approval Workflow	As a manager, I want to receive, review, and approve or reject leave requests so that I can manage team attendance.	High
US-04	Half-Day Leave Support	As an employee, I want to select "1st Half" or "2nd Half" leave so that I don't have to use a full day's credit for short absences.	Medium
US-05	Leave Tracking Dashboard	As a user, I want a dashboard to track my pending requests and remaining balance so that I can plan my time off.	High
US-06	Team Notifications	As a team member, I want to be notified when a colleague's leave is approved so that I can coordinate work coverage.	Medium

US-07	Annual Holiday Calendar / Holiday Overlap Logic	As an employee, I want to view the company holiday list so that I can align my personal leave with public holidays.	Low
US-08	Leave Cancellation	As an employee, I want to cancel a request before it is approved so that my balance is not deducted by mistake.	Medium

7.2 Sprint and Sprint Backlog

Sprint Duration: 2 weeks

Sprint Goal: Enable employees to submit full/half-day leave requests with reasons and allow managers to approve or reject them.

7.2.1 Sprint 1(Building core foundation):

Goal: Enable the basic "Submit to Approval" flow for full and half-days.

Sprint Backlog Tasks:

- Build User Authentication
- Develop basic Leave Application
- Implement Half day Logic and Full day logic(along with deduction)
- Create Dashboard for “Pending Approval”
- Updating Status Change (Pending → Approved/Rejected)

7.2.2 Sprint 2(Visibility & Communication):

Goal: Build the tracking dashboards and automated notifications for the team.

Sprint Backlog Tasks:

- Build Employee Dashboard (Visual cards for Sick, Holiday, Maternity balances).
- Create a Leave History table.
- Integrate Email/Notification Service (Triggered on Manager Approval).
- Build Team Calendar View (For seeing who is out today/this week).

7.2.3 Sprint 3(Validation):

Goal: Preventing double booking and handling holidays.

Sprint Backlog Tasks:

- Implement Overlap Validation (Block request if date is already booked).
- Holiday Calendar Integration (System ignores public holidays in day-count).
- Cancellation Workflow (Allow employees to withdraw "Pending" requests).

7.2.4 Sprint 4(Final touches):

Goal: User management, reporting, and production readiness.

Sprint Backlog Tasks:

- Add/Remove Employees and set their initial leave balances (by admins only).
- Export Reports (CSV/PDF) for HR payroll processing.
- Audit Logs (Track who approved what and when for compliance).
- Final UAT (User Acceptance Testing) and Bug Squashing.

7.3 Story point

Story ID	Requirement	Complexity Factor	Story Points
US-01	Leave Type Selection	Medium: Requires frontend dropdown and backend DB schema updates.	3
US-02	Compulsory Reason	Low: Simple form validation logic on the frontend and backend.	2
US-03	Manager Approval Workflow	High: Complex state-machine logic involving multiple user roles and security.	8
US-04	Half-Day Leave Support	Medium High: Requires custom date time logic and balance calculations.	5
US-05	Leave Tracking Dashboard	Medium: Requires data aggregation from multiple tables and a clean UI layout.	5

US-06	Team Notifications	Medium: Requires integration with an email service or internal notification system.	3
US-07	Annual Holiday Calendar / Holiday Overlap Logic	Low: Simple retrieval of a static or semi-static list of dates.	2
US-08	Leave Cancellation	Medium: Requires status rollback logic and balance reconciliation.	3

7.4 Retrospective

Sprint 1 Retrospective

Sprint Goal: Core Leave Workflow (Apply & Approve)

Date: [End of each Sprint]

Category	Observations & Feedback
What went well?	The "Half-day" logic was finished ahead of schedule. Team communication on the Database schema was very clear. Zero bugs found in the Leave Type dropdown.
What didn't go well?	The "Manager Approval" UI took longer than expected because of unclear CSS requirements. We forgot to account for what happens if a manager is <i>also</i> on leave (Delegation).
Improvements for Sprint 2	Define UI mockups before the sprint starts to avoid CSS delays. Add "Manager Delegation" to the Product Backlog for Sprint 4.

The sprint life cycle is repeated as:

Product Backlog created → Sprint Planning conducted → Sprint Backlog defined → Development & Testing executed → Effort Estimation(Story Point)→ Retrospective meeting held → Cycle repeats

8. Acceptance Criteria

Acceptance Criteria define the exact conditions that must be met for a User Story to be marked as "Done."

US-01: Leave Type Selection

- The leave application form must include a dropdown menu for "Leave Type."
- The dropdown must include at least the following options: Sick, Holiday, and Maternity.
- The system must allow the user to select only one leave type per application.
- The selected leave type must be successfully saved to the database upon submission.

US-02: Compulsory Reason

- The leave application form must include a text area field labeled "Reason."
- If the "Reason" field is left empty, the UI must display a clear error message stating: "Reason is mandatory."
- The submitted reason must be visible to the manager during the review process.

US-03: Manager Approval Workflow

- Managers must have a dedicated view/dashboard showing a list of all "Pending" requests from their direct reports.
- Managers must be able to click "Approve" or "Reject" on any pending request.
- Clicking "Approve" must change the request status to "Approved" and officially deduct the leave balance.
- Clicking "Reject" must change the status to "Rejected" and require the manager to input a rejection comment.

US-04: Half-Day Leave Support

- The UI must display a toggle or checkbox allowing the user to select "Half-Day."
- When "Half-Day" is selected, the user must be forced to choose either "1st Half" or "2nd Half."

- The system must calculate the leave duration as 0.5 days.
- Approved half-day requests must deduct exactly 0.5 days from the employee's total balance.

US-05: Leave Tracking Dashboard

- Employees must have a dashboard displaying their leave statistics.
- The dashboard must show three distinct numbers for each leave type: Total Allowed, Used, and Remaining Balance.
- The "Used" and "Remaining Balance" calculations must only factor in "Approved" leave requests, not "Pending" or "Rejected" ones.

US-06: Team Notifications

- The system must automatically trigger a notification when a manager clicks "Approve."
- The notification must be sent to the designated team channel (e.g., Email group or Slack/Teams channel).
- The message must include the employee's name, the dates of the leave, and specify if it is a half-day.

US-07: Annual Holiday Calendar

- Employees must have access to a read-only page or widget displaying the company's official holiday list.
- The list must be organized chronologically by date.
- The system must automatically recognize these dates and not deduct from an employee's leave balance if their requested date range overlaps with a public holiday.

US-08: Leave Cancellation

- Employees must see a "Cancel" button next to any leave request currently in the "Pending" status.
- Clicking "Cancel" must change the status to "Cancelled" and remove it from the manager's approval queue.
- The system must not deduct any leave balance for cancelled requests.

9. Testing Criteria

Testing to be conducted are:

- Manual Testing
- Automated Testing
- API Testing
- Smoke Testing
- Regression Testing
- White Box Testing
- Black Box Testing
- Non-Functional Testing
- User Acceptance Testing

10. RTM(Requirement Traceability Matrix)

Req ID	Requirement Description	User Story ID	Sprint	Test Case ID
REQ-01	Employees must select a specific leave type.	US-01	Sprint 1	TC-01, TC-08
REQ-02	A text reason is mandatory for all applications.	US-02	Sprint 1	TC-03
REQ-03	Requests must be routed to managers for approval.	US-03	Sprint 1	TC-04, TC-05, TC-11
REQ-04	System must support 1st/2nd Half-Day leaves.	US-04	Sprint 1	TC-02
REQ-05	Employees must be able to track their balances.	US-05	Sprint 2	TC-06
REQ-06	System must automatically notify the team upon approval.	US-06	Sprint 2	TC-07, TC-12

REQ-07	System must handle Public Holidays correctly.	US-07	Sprint 3	TC-10
REQ-08	Employees can cancel unapproved leave requests.	US-08	Sprint 3	TC-09